

WASP-36b

R-0661

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_260320 · created 2026-07-27T17:39:35+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461119.7386 +/- 0.0063 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.048
Transit depth (Rp/Rs)^2	1.98 +/- 1.35 [%]
Orbital Inclination (inc)	80.18 +/- 2.9
Airmass coefficient 1 (a1)	1.53 +/- 0.91
Airmass coefficient 2 (a2)	0.03 +/- 0.76
Scatter in the residuals of the lightcurve fit is	63.16 %
Best Comparison Star	None
Optimal Aperture	2.17
Optimal Annulus	7.5
Transit Duration (day)	0.039 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.141 ± 0.048 vs 0.1368 (deviation 0.09σ)
Duration fit vs published	0.039 d vs 0.0758 d (deviation 1.12σ)
Mid-time O–C	15.7 min
Depth SNR	2.9
Residual scatter	63.16 %

Light curve

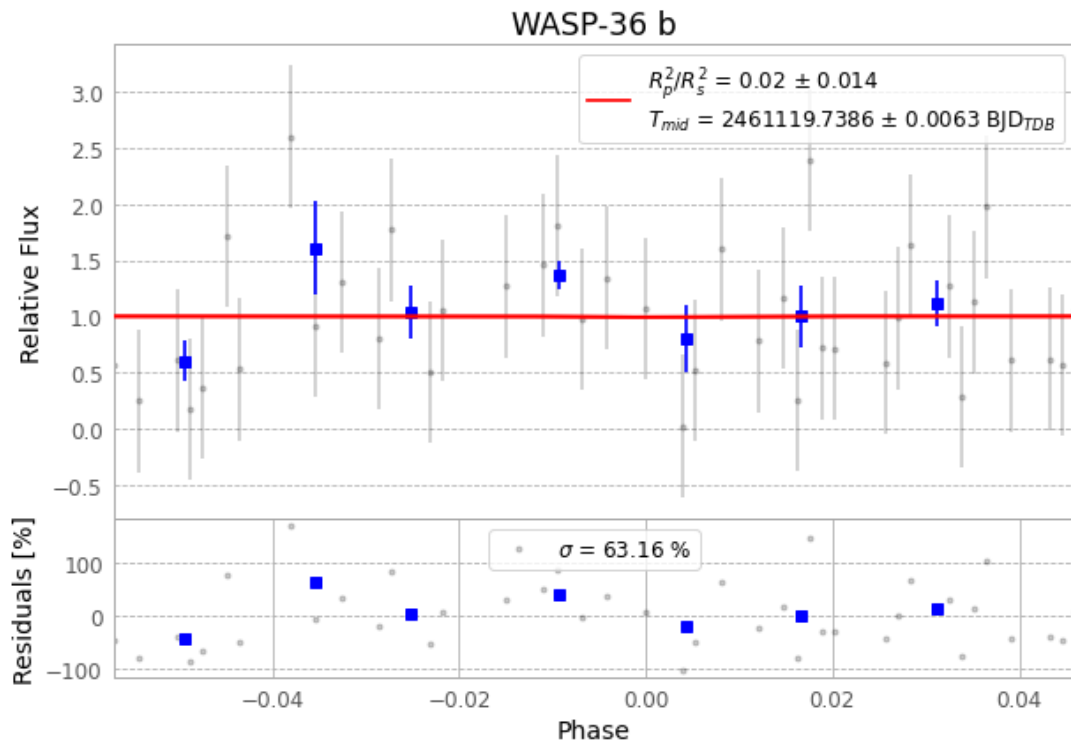


Figure 1 — FinalLightCurve_WASP-36 b_2026-03-19.png (SHA-256 2b6eaf16367217bf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [523, 302], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d5c38a9aab4b82d1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 574e53e

Data provenance

77 FITS frames (51 MB), WASP-36260320033015.FITS → WASP-36260320071815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-260320.log (17f1072c1f53...), FinalLightCurve_WASP-36 b_2026-03-19.png (2b6eaf163672...), FinalLightCurve_WASP-36 b_2026-03-19.csv (946af4203e7a...)

Compute resources

Wall time 445.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 17:39Z.